

POOJAN AKHANI

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EDUCATION

Master of Science, Computer Science University at Albany, SUNY Relevant Coursework: Artificial Intelligence, Database Systems, Operating Systems, Software Engineering, Algorithms & Data Structures, Probability and Computing	Aug 2023 - May 2025 GPA: 3.75/4.0
Bachelor of Engineering, Computer Engineering Gujarat Technological University, India	Jun 2019 - May 2023 GPA: 3.5/4.0

SKILLS

Programming Languages & Scripting: Java, Python, C++, C, C#, PHP, Kotlin, Dart, JavaScript, TypeScript, VB.NET
Web & Application Development: React, Angular, Node.js, FastAPI, Spring Boot, ASP.NET, Hibernate, HTML5/CSS3
Cloud & DevOps Tools: AWS, Microsoft Azure, Google Cloud Platform (GCP), Git, GitHub, Docker, Kubernetes, Jenkins, Maven, Postman, JIRA
Databases & Data Engineering: MySQL, PostgreSQL, MongoDB, Firebase, Oracle, PL/SQL, DynamoDB, Aurora, ETL, Data Warehousing
Analytics, AI & Methodologies: Power BI, Tableau, Excel, IBM Cognos, OBIEE, TensorFlow, PyTorch, Keras, Scikit-learn, Agile, Scrum, TDD

EXPERIENCE

Student Assistant Department of Computer Science, University at Albany, SUNY	Albany, USA Aug 2024 - May 2025
- Designed and implemented an automated grading system in Python to evaluate Java-based student assignments by extracting and executing core program files against predefined test cases, improving evaluation accuracy by 40% and cutting manual grading time by 50%. Gathered BI requirements to support curriculum-level insights and academic planning. - Developed interactive dashboards and BI reports using Tableau and Power BI to visualize key academic performance indicators. Collaborated with faculty in an Agile environment to align instructional content with educational standards, resulting in a 30% boost in student outcomes and a 25% improvement in curriculum alignment.	
Software Developer Zetrixweb Infotech LLP	Ahmedabad, India Jul 2022 - Jun 2023
- Built full-stack applications using Java (Spring Boot), React.js, TypeScript, and MySQL, deploying on AWS (EC2, S3, RDS) and Firebase to deliver secure, scalable systems with 30% faster response times. - Integrated machine learning models with Python, scikit-learn, Pandas, NumPy, Flask, and REST APIs, adding predictive analytics that improved decision accuracy by 25%. - Automated CI/CD workflows using Jenkins, Docker, Git, Kubernetes, and Jira, streamlining Agile releases and reducing deployment time by 20%.	
Java Developer Next Technolabs LLP	Ahmedabad, India May 2022 - Jun 2022
- Developed robust Java applications emphasizing modular design, exception handling, and backend integration, enhancing system reliability by 25% and performance by 18% through refined SQL queries and efficient data flow across layers. - Applied design patterns such as MVC and Singleton to scale applications effectively, resulting in a 20% boost in responsiveness and maintainability. Leveraged tools like JDBC, Git, and IntelliJ IDEA to streamline development and version control in collaborative environments.	

PROJECTS

Game Recommendation System TypeScript, Angular, Python, Pandas, NumPy, FastAPI, Redis, AWS	Jan 2025 - May 2025
- Developed a full-stack game recommendation engine using Angular for the frontend and FastAPI with Python, Pandas, and NumPy for backend processing, implementing Redis to cache recommendations and reduce latency. The application is hosted on AWS infrastructure for high availability and scalability. - Leveraged Large Language Models (LLMs) to analyze game metadata, improving recommendation accuracy by 30% based on user preferences and behavior. Stored user profiles and favorites in AWS RDS for persistence and analytics.	
Retriever-Augmented Generation (RAG) Python, Transformers, FAISS, LangChain, Hugging Face	Aug 2024 - Dec 2024
- Formulated a RAG pipeline using LangChain, FAISS, and Hugging Face Transformers for efficient document retrieval and context-aware question answering on the TriviaQA dataset, reducing response time by 50% and answer relevance by 30%. - Pre-processed data, developed embedding-based retrievers, and fine-tuned models to enhance semantic search and response generation.	
UniChat Flutter, RESTful APIs, MongoDB, Firebase integrations Scrum Master	Jan 2024 - Apr 2024
- Integrated Discord, Google Docs, and OpenAI's ChatGPT to establish a cohesive academic collaboration platform, boosting user engagement and productivity. - Set up RESTful APIs to facilitate MongoDB integration, ensuring real-time data synchronization for efficient web application functionality and seamless user experience.	
E-Commerce Website Java, Hibernate, Spring Boot, MySQL, HTML, CSS, Bootstrap	Jun 2021 - Oct 2021
- Developed a full-stack e-commerce platform with product management, order processing, and secure payment integration, enhancing user experience. - Optimized responsiveness and performance using HTML, CSS, and JavaScript, ensuring a smooth and engaging shopping experience.	

CERTIFICATIONS

- AWS Certified Cloud Practitioner** | Amazon Web Services
- Java Project Trainee** | Tops Technologies

POOJAN RASIKLAL AKHANI

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Date: October 24, 2025

Hiring Committee
College of Engineering, Robotics Department
University of Michigan
Ann Arbor, MI

Subject: Application for Applications Programmer Associate (Job Opening ID: 270067)

Dear Hiring Committee,

I am writing to express my interest in the Applications Programmer Associate position within the College of Engineering's Robotics Department at the University of Michigan. With a strong foundation in software development, database management, and web application design, I am confident in my ability to support the department's mission of building innovative and people-centered technologies.

As a Software Developer at Zetrixweb Infotech, I developed and maintained full-stack web applications using Java Spring Boot, React, and MySQL. I designed RESTful APIs, optimized database queries, and automated deployments using Docker and Jenkins. I collaborated closely with cross-functional teams to implement system enhancements and troubleshoot issues across environments, ensuring high reliability and compliance with security standards. This experience directly aligns with the responsibilities of developing, testing, and maintaining software applications within the Robotics Department.

During my time as a Student Assistant at the University at Albany, I created Python scripts to automate data processing tasks and built Power BI dashboards for visual reporting. I worked with faculty to gather requirements, translate user needs into functional tools, and document workflows to support technical consistency and reproducibility. These experiences strengthened my skills in requirement analysis, data visualization, and user-centered development—essential qualities for this position.

My academic and technical projects demonstrate a blend of analytical and development expertise. I developed a Retriever-Augmented Generation pipeline using LangChain and FAISS to optimize information retrieval, and I built a game recommendation system using FastAPI, Redis, and AWS for scalable performance. These projects required extensive debugging, testing, and iterative collaboration, reflecting my commitment to quality assurance and efficient application design.

Key Qualifications:

- **Programming and Development:** Proficient in Python, JavaScript, and Java for building scalable applications and automation tools.
- **Web Technologies:** Experienced in React, Node.js, Flask, and FastAPI for modern web and backend development.
- **Database Management:** Skilled in SQL, MySQL, and PostgreSQL for query optimization and data integrity.
- **System Deployment:** Hands-on experience with Docker, AWS, and Jenkins for secure and automated deployment workflows.
- **Collaboration and Documentation:** Strong communicator with experience translating user needs into reliable, maintainable solutions.

I hold a Master of Science in Computer Science from the University at Albany, SUNY, and a Bachelor of Engineering in Computer Engineering. My technical experience, academic background, and collaborative mindset make me an excellent fit for the Robotics Department's programming and application development needs.

Thank you for considering my application. I would welcome the opportunity to discuss how my skills and experience can contribute to the innovative research and operational goals of the University of Michigan.

Sincerely,
Poojan Akhani